Continuous Probability Distributions Topic Test 1

1. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{1}{2}sinx & 0 < x < k \\ 0 & x \geq k \; or \; x \leq 0 \end{cases}$.

   What is the value of k?

$$\int_0^k \frac{1}{2}\sin x = 1$$

$$\left[-\frac{1}{2}\cos x\right]_0^k = 1$$

$$-\frac{1}{2}(\cos k - 1) = 1$$

$$\cos k - 1 = -2$$

$$\cos k = -1$$

$$\boxed{k = \pi}$$

2. The handspans of a large sample of 17-year-old students were measured and recorded. These handspans are approximately normally distributed with a mean of 16.2 cm and a standard deviation of 2.7 cm. A 17-year-old student has a handspan 13.9 cm. What is this student's standardised handspan (z score), correct to one decimal place?

$$z = \frac{x - \bar{x}}{s.d.}$$

$$z = \frac{13.9 - 16.2}{2.7}$$

$$\therefore \boxed{z = -0.9}$$

3. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{x^3}{4} & 0 \leq x \leq 2 \\ 0 & x > 2 \; or \; x < 0 \end{cases}$. The median of this probability density function is closest to:

   A. 1.26 B. 1.41 C. 1.50 D. 1.68

Cumulative Density Function (CDF):

$$F(x) = \int_0^x \frac{x^3}{4}\,dx$$

$$= \left[\frac{x^4}{16}\right]_0^x$$

$$F(x) = \frac{x^4}{16}$$

$$\frac{1}{2} = \frac{x^4}{16}$$

$$x^4 = 8$$

$$\therefore \boxed{x = 1.68}$$

4. Kerry's class achieved a 72% mean and 8% standard deviation for their project work. What was Kerry's mark if he achieved a z-score of -2.5?

$$z = \frac{x - \bar{x}}{s.d.}$$

$$-2.5 = \frac{x - 72}{8}$$

$$-20 = x - 72$$

$$\therefore \boxed{x = 52\%}$$

Continuous Probability Distributions Topic Test 1

5. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{3}{4}(x^2-1) & 1 < x < 2 \\ 0 & x \le 1 \text{ or } x \ge 2 \end{cases}$.

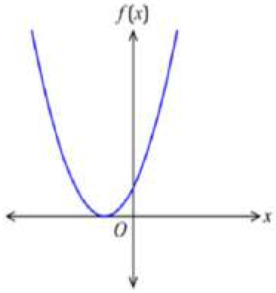
The value of $P(X \le 1.3)$ is closest to:

A. 0.07 B. 0.30 C. 0.43 D. 0.93

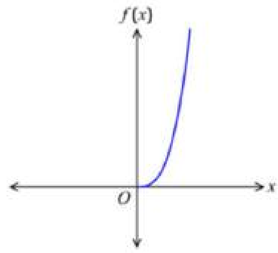
$$P(X \le 1.3) = \int_0^1 0\,dx + \int_1^{1.3} \frac{3}{4}(x^2-1)\,dx$$
$$= 0 + \frac{3}{4}\left[\frac{x^3}{3} - x\right]_1^{1.3}$$
$$= \frac{3}{4}\left[\left(\frac{1.3^3}{3} - 1.3\right) - \left(\frac{1^3}{3} - 1\right)\right]$$
$$= \boxed{0.07}$$

6. Which of the following gra[...] ty function $f(x)$?

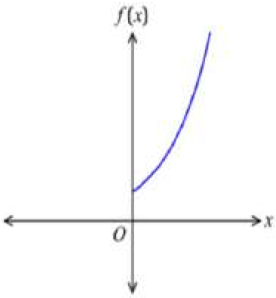
A.



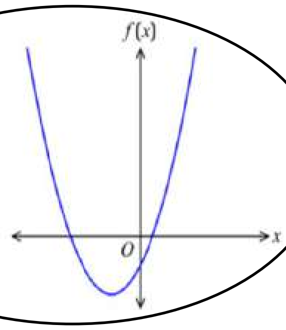
B.



C.



D.



7. Results for a class test are given as z-scores. In this test, Levi gained a z-score equal to -2. Which of the following is the correct interpretation of his z-score in terms of the mean and standard deviation?

A. Levi's score was 2 means below the standard deviation.
B. Levi's score was 2 standard deviations below the mean.
C. Levi's score was 2 standard deviations above the mean.
D. Levi's score was 2 means deviation above the standard deviation.

8. The probability density function for the continuous random variable X is:

$$f(x) = \begin{cases} kx^3 + \frac{3}{4}x & 0 < x < 2 \\ 0 & otherwise \end{cases}$$

What is the value of k?

$$\int_0^2 kx^3 + \frac{3}{4}x \, dx = 1$$
$$\left[\frac{kx^4}{4} + \frac{3x^2}{8}\right]_0^2 = 1$$
$$\left(\frac{k(2)^4}{4} + \frac{3(2)^2}{8}\right) - (0) = 1$$
$$4k + \frac{3}{2} = 1$$
$$4k = -\frac{1}{2}$$
$$\boxed{\therefore k = -\frac{1}{8}}$$

9. A class of students sat for a Mathematics test and an Ancient history test. Each test had a possible maximum score of 100 marks. The table below shows the mean and standard deviation of the marks obtained in these tests.

	Mathematics	Ancient history
Class mean	60	68
Class standard deviation	12	8

The class marks in each subject are approximately normally distributed. Omar obtained a mark of 81 in the Mathematics test. What mark would Omar need to obtain in the Ancient history test to achieve the same standardised score as he did for Mathematics?

$z = \frac{x - \bar{x}}{s.d.}$

Maths : $z = \frac{81 - 60}{12} = \frac{7}{4}$

Ancient History : $\frac{7}{4} = \frac{x - 68}{8}$

$14 = x - 68$

$\therefore \boxed{x = 82}$

10. The probability density funct[illegible] variable X is: $f(x) = \begin{cases} \frac{3}{2}(x-1)(x-2)^2 & 1 \le x \le 3 \\ 0 & x < 1 \text{ or } x > 3 \end{cases}$.

What is mean or expected va[illegible]

$\text{Mean} = \int_a^b x f(x)\, dx$

$= \int_1^3 x \times \frac{3}{2}(x-1)(x-2)^2\, dx$

$= \frac{3}{2}\int_1^3 (x^2 - x)(x^2 - 4x + 4)\, dx$

$= \frac{3}{2}\int_1^3 x^4 - 4x^3 + 4x^2 - x^3 + 4x^2 - 4x\, dx$

$= \frac{3}{2}\int_1^3 x^4 - 5x^3 + 8x^2 - 4x\, dx$

$= \frac{3}{2}\left[\frac{x^5}{5} - \frac{5x^4}{4} + \frac{8x^3}{3} - \frac{4x^2}{2}\right]_1^3$

$= \boxed{\frac{13}{5} \text{ or } 2.6 \text{ or } 2\frac{3}{5}}$

11. The probability densi[illegible] X is: $f(x) = \begin{cases} \frac{8}{3}(1-x) & 0 < x < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$.

The median of this pr[illegible]

A. 0.22　　B. 0.21　　C. 0.67　　D. 1.79

CDF :

$F(x) = \int_0^x \frac{8}{3}(1-x)\, dx$

$\frac{1}{2} = \frac{8}{3}\left[x - \frac{x^2}{2}\right]_0^x$

$\frac{1}{2} = \frac{8}{3}\left(x - \frac{x^2}{2}\right)$

$\frac{3}{16} = x - \frac{x^2}{2}$

$3 = 16x - 8x^2$

$8x^2 - 16x + 3 = 0$

$x = 0.21$ or 1.79

but $0 < x < \frac{1}{2}$

$\therefore \boxed{x = 0.21}$

12. Thomas scored 76 in a Mathematics test. The Mathematics test had a mean of 60 and a standard deviation of 8. A Modern history test had a mean of 56 and a standard deviation of 12. What mark in the History test would have been equivalent to Thomas's mathematics mark?

$z = \frac{x - \bar{x}}{s.d.}$

Maths: $z = \frac{76 - 60}{8} = 2$

History: $2 = \frac{x - 56}{12}$

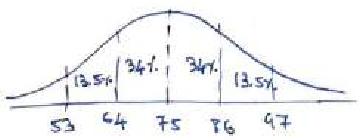
$24 = x - 56$

$\therefore \boxed{x = 80}$

13. Isabella scored 81 in an asses … was 67 with a standard deviation of 7.0. What is Isabella's z-score?

$z = \frac{x - \bar{x}}{s.d.} = \frac{81 - 67}{7} = \boxed{2}$

14. The pulse rates of a large g … lly distributed with a mean of 75 beats/minute and a standa … 18-year-old students could be expected to have pulse rates greater than 53 beats/minute but less than 86 beats/minute?



Total = 13.5% + 34% × 2

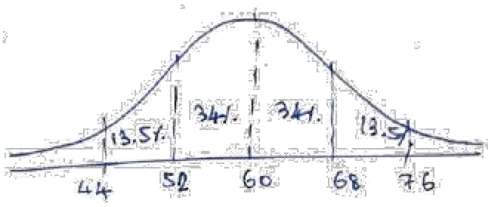
$\boxed{\text{Total} = 81.5\%}$

15. The probability density function … le X is: $f(x) = \begin{cases} 2(1 - \frac{1}{x^2}) & 1 \le x \le 2 \\ 0 & otherwise \end{cases}$.

What is mean or expected value of X?

$$\text{Mean} = \int_a^b x f(x)\, dx$$
$$= \int_1^2 x \times 2(1 - \tfrac{1}{x^2})\, dx$$
$$= \int_1^2 2x - \tfrac{2}{x}\, dx$$
$$= \left[x^2 - 2\ln x\right]_1^2$$
$$= 4 - 2\ln 2 - 1$$
$$= \boxed{3 - 2\ln 2 \text{ or } 1.61}$$

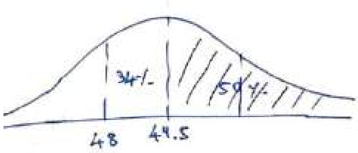
16. In a normally distributed set of scores, the … n is 8. Approximately what percentage of the scores will lie between 44 and 76?



Total = $(13.5 + 34) \times 2 = \boxed{95\%}$

17. The head circumference (in cm) of a population of infants is normally distributed with a mean of 49.5 cm and a standard deviation of 1.5 cm. Five hundred of these infants are selected at random and each infant's head circumference is measured. What is the expected number of these infants with a head circumference of less than 48.0 cm?

A. 12 B. 13 C. 80 D. 420



Less than 48 cm

= 100 − 50 − 34

= 16%

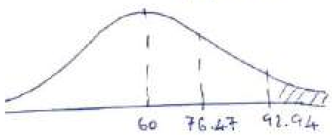
16% × 500 = $\boxed{80 \text{ infants}}$

18. Liam's class achieved the fol... 4, 46, 73, 63, 57 and 29

How many scores have a z-score greater than 2?

1. Using the calculator:

$\bar{x} = 60$, s.d. = 16.47 (population)



z-score greater than 2

= $\boxed{1 \text{ person}}$

19. If the function $f(x) = 4x$ represents a probability d... he following could be the domain of $f(x)$?

A. $0 \le x \le \frac{1}{\sqrt{2}}$ B. $0 \le x \le 0.25$

C. $0 \le x \le 0.5$ D. $0 \le x \le 1$

$\int_0^k 4x \, dx = 1$

$\left[\frac{4x^2}{2}\right]_0^k = 1$

$2k^2 = 1$

$k^2 = \frac{1}{2}$

$\therefore \boxed{k = \frac{1}{\sqrt{2}}}, \quad k > 0$

$\therefore \boxed{A}$

20. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{1}{9}(4x - x^2) & 0 < x < 3 \\ 0 & otherwise \end{cases}$.

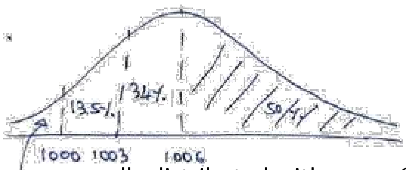
Find the value of $P(X \le 2)$.

$P(X \le 2) = \int_0^2 \frac{1}{9}(4x - x^2) \, dx$

$= \frac{1}{9}\left[2x^2 - \frac{x^3}{3}\right]_0^2$

$= \boxed{\frac{16}{27} \text{ or } 0.59}$

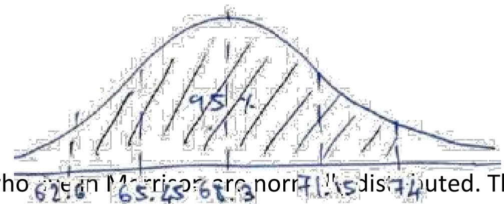
21. The weight of sugar sold in packets is known to have a bell-shaped distribution with a mean of 1006 grams and a standard deviation of 3 grams. What percentage of packets of sugar would be expected to contain less than 1000 grams?



$100\% - 50\% - 34\% = 13.5\%$

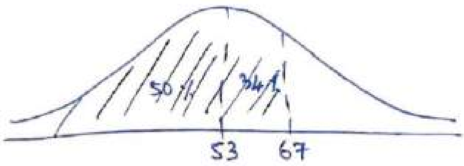
$= \boxed{2.5\%}$

22. The heights of standard poodles are normally distributed with mean 68.3 cm and standard deviation 2.85 cm. Between which two values would it be expected that the heights of 95% of standard poodles lie?



Between $= \boxed{62.6 \text{ and } 74}$

23. The ages of the citizens who live in [illegible] are normally distributed. The mean age is 53 years and the standard deviation is 14. What percentage of the citizens is younger than 67?



Younger than 67 $= 50 + 34$

$= \boxed{84\%}$

24. The probability density function [illegible] is: $f(x) = \begin{cases} \frac{3}{4}(x^2-1) & 1 < x < 2 \\ 0 & x \le 1 \text{ or } x \ge 2 \end{cases}$.

The mean $E(X)$ is equal to:

A. 1.50 ... $\frac{}{32}$ D. $\frac{27}{16}$

$E(x) = \int_a^b x f(x)\, dx$

$= \int_1^2 x \times \frac{3}{4}(x^2-1)\, dx$

$= \frac{3}{4}\int_1^2 x^3 - x \, dx$

$= \frac{3}{4}\left[\frac{x^4}{4} - \frac{x^2}{2}\right]_1^2$

$= \boxed{\frac{27}{16} \text{ or } 1\frac{11}{16} \text{ or } 1.6875}$

Continuous Probability Distributions Topic Test 2

1. A continuous random variable x has a probability density function f given $f(x) = \begin{cases} Ax+b & 1 \le x \le 6 \\ 0 & elsewhere \end{cases}$ where A and B are constants. The median of x is 3. Find the values of A and B.

X is a random variable $1 \le x \le 6$,

$$\therefore \int_1^6 (Ax+B)dx = 1$$

$$\left[\frac{Ax^2}{2} + Bx\right]_1^6 = 1$$

$$\frac{35A}{2} + 5B = 1$$

$$35A + 10B = 2 \quad (1)$$

Also $\int_1^3 (Ax+B)dx = \frac{1}{2}$ (Median 3)

$$\left[\frac{Ax^2}{2} + Bx\right]_1^3 = \frac{1}{2}$$

$$4A + 2B = \frac{1}{2} \quad (2)$$

Multiply equation (2) by 5.

$$20A + 10B = \frac{5}{2} \quad (3)$$

Equation (1) – Equation (3)

$$15\text{A} = -\frac{1}{2}$$

$$A = -\frac{1}{30}$$

Substitute $A = -\frac{1}{30}$ in equation (1).

$$35 \times -\frac{1}{30} + 10B = 2$$

$$B = \frac{19}{60}$$

$$\therefore A = -\frac{1}{30} \text{ and } B = \frac{19}{60}$$

2. A continuous random variable x has a probability density function f given $f(x) = \begin{cases} k\sin(\pi x) & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$. Show that $k = \frac{\pi}{2}$.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

$$\int_0^1 k\sin(kx)dx = 1$$

$$\left[\frac{-k\cos(\pi x)}{\pi}\right]_0^1 = 1$$

$$\frac{-k\cos\pi}{\pi} + \frac{k\cos 0}{\pi} = 1$$

$$\frac{k}{\pi} + \frac{k}{\pi} = 1$$

$$\frac{2k}{\pi} = 1$$

$$\therefore k = \frac{\pi}{2}$$

3. The speed limit outside a school is 40 km/h. Year 11 students measured the speed of passing vehicles over a period of time. They found the set of data to be normally distributed with a mean speed of 36 km/h and a standard deviation of 2 km/h. What percentage of the vehicles passes the school at a speed greater than 40 km/h?

$$z = \frac{x-\mu}{\sigma} = \frac{40-36}{2} = 2$$

z-score greater than 2.

$$\text{Percentage} = (100\% - 95\%) \div 2$$
$$= 2.5\%$$

4. Let X be a normally distributed random variable with mean of 5 and variance of 9. Also let Z be the random variable with the standard normal distribution.

a. Find $P(X > 5)$.

b. Find b such that $P(X > 7) = P(Z < b)$.

(a)

$\sigma^2 = 9$ (or $\sigma = 3$) and $\mu = 5$

$P(X > 5) = P(X > \mu) = 0.5$

(b)

$P(X > 7) = P(X < 3)$ (symmetry)

$$z = \frac{x-\mu}{\sigma} = \frac{3-5}{3} = -\frac{2}{3}$$

$$P(X > 7) = P\left(Z < -\frac{2}{3}\right)$$

$$\therefore b = -\frac{2}{3}$$

Continuous Probability Distributions Topic Test 2

5. A continuous random variable X has a probability density function f given $f(x) = \begin{cases} ax(5-x) & 0 \le x \le 5 \\ 0 & elsewhere \end{cases}$ where a is a positive constant.
   a. Find the value of a.
   b. Express $P(X < 3)$ as a definite integral.

(a)

$$\int_0^5 ax(5-x)dx = 1$$
$$\int_0^5 5ax - ax^2\, dx = 1$$
$$\left[\frac{5ax^2}{2} - \frac{ax^3}{3}\right]_0^5 = 1$$
$$\frac{125a}{2} - \frac{125a}{3} = 1$$
$$a = \frac{6}{125}$$

(b)

$$P(X < 3) = \int_0^3 \frac{6}{125} x(5-x)dx$$

6. The heights of Year 12 girls are normally distributed with a mean of 165 cm and a standard deviation of 5.5 cm. What is the $z-$score for a height of 154 cm?

$$z = \frac{x-\mu}{\sigma} = \frac{154-165}{5.5} = -2$$

7. A continuous random variable X has a probability density function f given by $f(x) = \begin{cases} \cos(2x) & \frac{3\pi}{4} < x < \frac{5\pi}{4} \\ 0 & otherwise \end{cases}$.
   Find the value of a such that $P(X < a) = 0.25$, correct your answer to 2 decimal places.

$$\int_{\frac{3\pi}{4}}^{a} \cos 2x dx = 0.25$$
$$[0.5 \sin 2x]_{\frac{3\pi}{4}}^{a} = 0.25$$
$$0.5\left(\sin 2a - \sin\frac{3\pi}{2}\right) = 0.25$$
$$\sin 2a + 1 = 0.5$$
$$2a = \frac{11\pi}{6}$$
$$a \approx 2.88$$

8. A continuous random variable X has a function f given by $f(x) = \begin{cases} kx & if\ 0 \le x \le 2 \\ 0 & if\ x > 2\ or\ x < 0 \end{cases}$ where k is a constant.
   a. Find the value of k that makes f a probability density function.
   b. Find $P(X > 1.5)$.

(a)

Probability density function

$$\int_{-\infty}^{\infty} f(x)dx = 1$$
$$\int_0^2 kx\, dx = 1 \quad (f(x) = 0 \text{ elsewhere})$$
$$\left[\frac{kx^2}{2}\right]_0^2 = 1$$
$$2k = 1$$
$$k = 0.5$$

(b)

$P(X > 1.5)$

$$\int_{1.5}^2 0.5x\, dx = \left[\frac{x^2}{4}\right]_{1.5}^2$$
$$= \frac{2^2}{4} - \frac{1.5^2}{4}$$
$$= 0.4375$$

9. The weights of 10 000 newborn babies in NSW are normally distributed. These weights have a mean of 3.1 kg and a standard deviation of 0.35 kg. How many of these newborn babies have a weight between 2.75 kg and 4.15 kg?

A. 4985 B. 6570 C. 8370 D. 8385

(D)

$$z = \frac{x-\mu}{\sigma} = \frac{2.75-3.1}{0.35} = -1$$

$$z = \frac{x-\mu}{\sigma} = \frac{4.15-3.1}{0.35} = 3$$

$$\text{Percentage} = \frac{68\%}{2} + \frac{99.7\%}{2} = 83.85\%$$

$$\text{Newborns} = 83.85\% \times 10\,000 = 8385$$

10. In a normally distributed set of scores, the mean is 60 and a standard deviation is 8. Approximately what percentage of the scores will lie between 44 and 76?

$$z = \frac{x-\mu}{\sigma} = \frac{44-60}{8} = -2$$

$$z = \frac{x-\mu}{\sigma} = \frac{76-60}{8} = 2$$

95% of scores between -2 and 2

11. The diastolic measurement for blood pressure is 40-year-old people is normally distributed, with a mean of 80 and standard deviation of 20.

a. Calculate the $z-$ score for a diastolic measurement of 55.

b. The probability that a 50-year-old person has a diastolic measurement for blood pressure between 55 and 60 can be found by evaluating $\int_a^b f(x)\ dx$ where a and b are constants and where $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ is the normal probability density function with mean 0 and standard deviation 1. By first finding the values of a and b, calculate an approximate value for this probability by using the trapezoidal rule with three function values.

(a)

$$z = \frac{x-\mu}{\sigma} = \frac{55-80}{20} = -1.25$$

(b)

Three function values: 55, 57.5, 60

z-scores are the x values.

$x = -1.25$ (for 55)

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-(-1\cdot25)^2}{2}} = 0.1826\ldots$$

$x = -1.125$ (57.5)

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-(-1\cdot125)^2}{2}} = 0.2118\ldots$$

$x = -1$ (60)

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{\frac{-(-1)^2}{2}} = 0.2419\ldots$$

$$A = \frac{h}{2}(y_0 + 2y_1 + y_2)$$

$$= \frac{0.125}{2}(0.1826 + 2\times 0.2118 + 0.2420)$$

$$\approx 0.0530$$

$$\approx 5.30\%$$

12. The continuous random variable X has a normal distribution with mean 20 and standard devation 6. The continuous random variable Z has the standard normal distribution. The probability that Z is between -2 and 1 is equal to:

A. $P(8 < X < 32)$ B. $P(8 < X < 26)$

C. $P(14 < X < 26)$ D. $P(14 < X < 32)$

(B)

$$x = \mu + z\times\sigma$$

$$= 20 + (-2)\times 6 = 8$$

$$x = \mu + z\times\sigma$$

$$= 20 + (1)\times 6 = 26$$

$$\therefore P(-2 < Z < 1) = P(8 < X < 26)$$

Continuous Probability Distributions Topic Test 2

13. A machine produces cylindrical pipes. The mean of the diameters of the pipes if 8 cm and the standard deviation of 0.04 cm. Assuming a normal distribution, what percentage of cylindrical pipes produced will have a diameter less than 7.96 cm?

A. 16% B. 32% C. 34% D. 68%

(A) 7.96 is 1 standard deviation below the mean (as 8 – 7.96 = 0.04).
Percentage = 50% − 34%
= 16%

14. Kurt scored 81 in an assessment task. The mean for this task was 67 with a standard deviation of 7.0. What is Kurt's $z-$score?

$$z = \frac{x-\mu}{\sigma} = \frac{81-67}{7} = 2$$

Kurt's z-score is 2.

15. The results of two tests are normally distributed. The mean and standard deviation for each test are displayed in the table below.

	Mathematics	*English*
μ	70	75
σ	6.5	8

Kristoff scored 74 in Mathematics and 80 in English. He claims that he has performed better in English. Is Kristoff correct? Justify your answer using appropriate calculations.

Mathematics:
$$z = \frac{x-\mu}{\sigma} = \frac{74-70}{6.5} = 0.6154$$
English:
$$z = \frac{x-\mu}{\sigma} = \frac{80-75}{8} = 0.625$$
∴ Kristoff is correct (0.625 > 0.6154).

16. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{5}{x^2} & x \geq 5 \\ 0 & x < 5 \end{cases}$.

a. Find the probability that X is less than 12.
b. Find the probability that X is between 8 and 10.

(a)
$P(X > 12)$
$$\int_5^{12} \frac{5}{x^2}dx = \left[-\frac{5}{x}\right]_5^{12} = -\frac{5}{12} + 1 = \frac{7}{12}$$

(b)
$$\int_8^{10} \frac{5}{x^2}dx = \left[-\frac{5}{x}\right]_8^{10} = -\frac{1}{2} + \frac{5}{8} = \frac{1}{8}$$

17. The continuous random variable X has a normal distribution with mean 14 and standard deviation 2. If the random variable Z has the standard normal distribution, then the probability that X is greater than 17 is equal to:

A. $P(Z > 2)$ B. $P(Z > 3)$ C. $P(Z < 1.5)$ D. $P(Z < -1.5)$

(D) $$z = \frac{x-\mu}{\sigma} = \frac{17-14}{2} = 1.5$$
$$P(X > 17) = P(Z > 1.5)$$
$$= P(Z < -1.5)$$

Continuous Probability Distributions Topic Test 2

18. A continuous random variable X has a probability density function f given $f(x) = \begin{cases} \pi sin(2\pi x) & 0 \leq x \leq \frac{1}{2} \\ 0 & otherwise \end{cases}$. Find the value of a such that $P(X > a) = 0.2$, correct your answer to 2 decimal places.

$$\int_a^{0\cdot5} \pi\sin2\pi x dx = 0.2$$
$$[-0.5\cos 2\pi x]_a^{0\cdot5} = 0.2$$
$$-0.5(\cos\pi + \cos 2a\pi) = 0.2$$
$$0.5\cos 2a\pi = -0.3$$
$$\cos 2a\pi = -0.6$$
$$2a\pi \approx 2.2143\ldots$$
$$a \approx 0.35$$

19. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{1}{4} & 0 < x < 4 \\ 0 & x \leq 0 \; or \; x \geq 4 \end{cases}$.
    a. Find $F(x)$ the cumulative distribution of function X.
    b. Hence find $P(X \leq 3)$.

(a)

$$F(x) = \begin{cases} 0 & x \leq 0 \\ \frac{1}{4}x & 0 < x < 4 \\ 1 & x \geq 4 \end{cases}$$

(b)

$$\int_0^3 \frac{1}{4}xdx = \left[\frac{1}{4}x\right]_0^3$$
$$= \frac{3}{4}$$

20. A continuous random variable X has a function f given by $f(x) = \begin{cases} 6x(1-x) & 0 \leq x \leq 1 \\ 0 & otherwise \end{cases}$.
    a. Check that $f(x)$ is a probability density function.
    b. Sketch the function.
    c. Find $P(0 \leq X \leq 1)$.
    d. Find $P(0.5 \leq X \leq 1)$.

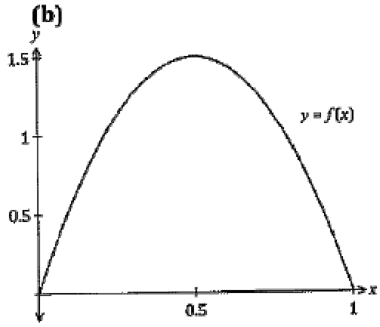
(a)

$f(x) = 6x(1-x) \geq 0$ in $0 \leq x \leq 1$
(Function is a quadratic with a negative coefficient and x-intercepts 0 and 1.)
$\therefore f(x) \geq 0$ for all x

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^1 6x(1-x)dx$$
$$= \int_0^1 6x - 6x^2\, dx$$
$$= [3x^2 - 2x^3]_0^1$$
$$= 1$$

$$\therefore \int_{-\infty}^{\infty} f(x)dx = 1$$

(b)



Probability density function

(c)

$P(0 \leq X \leq 1) = 1$
(See calculation in part (a).)

(d)

$P(0.5 \leq X \leq 1) = 0.5$
(Graph is symmetrical about $x = 0.5$.)

21. There are 60 000 students sitting a state-wide examination. If the results form a normal distribution, how many students would be expected to score a result between 1 and 2 standard deviations above the mean?
    A. 8 100 B. 16 200 C. 20 400 D. 28 500

(A) z-score between 1 and 2

$$\text{Percentage} = \frac{95\%}{2} - \frac{68\%}{2}$$
$$= 13.5\%$$
$$\text{Students} = 13.5\% \times 60\,000$$
$$= 8100$$

Continuous Probability Distributions Topic Test 2

22. Ali's class sits two Geography tests. The results of her class on the first Geography test are: 58, 74, 65, 66, 73, 71, 72, 74, 62, 70. The mean was 68.5 for the first test.
    a. Calculate the standard deviation for the first test. Give your answer correct to one decimal place.
    b. One the second Geography test, the mean for the class was 74.4 and the standard deviation was 12.4. Ali scored 62 in the first test. Calculate the mark that she needed to obtain in the second test to ensure that her performance relative to the class was maintained.

(a) Standard deviation
$\sigma = 5.5025\ldots$
≈ 5.5

(b) Ali's z-score on the first test.
$$z = \frac{x-\mu}{\sigma} = \frac{62-68.5}{5.5} = -1.1818\ldots$$
Ali maintains z-score on 2nd test.
$$x = \mu + z\times\sigma$$
$$= 74.4 + (-1.18)\times 12.4$$
$$= 59.8$$
Ali needs a mark of 59.8.

23. The probability density function for the continuous random variable X is: $f(x) = \begin{cases} \frac{1}{2}x & 0 \le x \le 2 \\ 0 & x < 0 \text{ or } x > 2 \end{cases}$. Find the mean or expected value of the random variable X.

$$E(x) = \int_{-\infty}^{\infty} xf(x)dx = \int_0^2 x\times\frac{1}{2}x\, dx$$
$$= \frac{1}{2}\int_0^2 x^2\, dx$$
$$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^2$$
$$= \frac{4}{3}$$

24. Sally's recent results in hospitality and timber are recorded in the table below.

Course	*Class mean*	*Class standard deviation*	*Sally's result*
Hospitality	55	10	85
Timber	55	15	85

a. What is Sally's $z-$ score for timber?
b. Explain the $z-$ score in timber in terms of the class mean and class standard deviation.
c. What hospitality mark would be equivalent to a $z-$ score of -1?

(a) $z = \frac{x-\mu}{\sigma} = \frac{85-55}{15} = 2$

(b) Sally has a z-score of 2, which is two standard deviations above the mean.

(c) $x = \mu + z\times\sigma$
$= 55 + (-1)\times 10$
$= 45$
The equivalent mark is 45.

25. Harry's class achieved the following results in a topic test: 39, 96, 78, 55, 65, 49, 61, 55, 74, 46, 73, 63, 57 and 29. How many scores have a $z-$ score greater than 2?

Mean is 60 and standard deviation is 16.5. A z-score greater than 2 would be 93 or higher. Therefore 96 is the only score greater than 93.

26. The probability density function for the continuous random variable x is: $f(x) = \begin{cases} 2(1-x) & 0 \le x \le 1 \\ 0 & x < 0 \text{ or } x > \end{cases}$. Find the median value of the random variable x.

Let m be the median.

$$\int_0^m 2(1-x)dx = 0.5$$

$$2\left[x - \frac{x^2}{2}\right]_0^m = 0.5$$

$$2m - m^2 = 0.5$$

$$m^2 - 2m + 0.5 = 0$$

$$\therefore m = 0.293 \text{ or } m = 1.707$$

But $0 \le x \le 1$

$\therefore m = 0.293$

27. The probability density function for the continuous random variable x is: $f(x) = \begin{cases} 2x^3 - x + 1 & 0 \le x \le 1 \\ 0 & otherwise \end{cases}$. Find $E(X)$.

To find the expected value or mean:

$$\int_{-\infty}^{\infty} xf(x)dx = \int_0^1 x(2x^3 - x + 1)\,dx$$

$$= \int_0^1 2x^4 - x^2 + x\,dx$$

$$= \left[\frac{2x^5}{5} - \frac{x^3}{3} + \frac{x^2}{2}\right]_0^1$$

$$= \frac{17}{30}$$